Multi-step methods

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Special multi-step methods ⇒

1. Explicit methods $y' = f(x,y)$, $y(x_0) = y_0$

Assuming that the solution of the above problem at $x_n, x_{n-1}, \dots x_{n-k+1}$ (k points), we want to find the solution at x_{n+1}.

x:	x_n	x_{n-1}	......	x_{n-k+1}
y_i:	y_n	y_{n-1}	......	y_{n-k+1}
$f_i = f(x_i, y_i)$	f_n	f_{n-1}	......	f_{n-k+1}

$y' = f(x,y) = \frac{dy}{dx}$. Now, integrate this wrt x ⇒

$$\int_{x_{n-l}}^{x_{n+1}} \frac{dy}{dx}\,.\,dx = \int_{x_{n-l}}^{x_{n+1}} f(x, y(x))\,dx \qquad \left[\begin{matrix} l \text{ is a non} \\ -\text{ve integer}\end{matrix}\right]$$

$$y(x_{n+1}) - y(x_{n-l}) = \int_{x_{n-l}}^{x_{n+1}} f(x, y(x))\,dx$$

Replace f by backward diff. interpolation polynomial interpolating at $x_n, x_{n-1}, \dots, x_{n-k+1}$. $[\nabla f_n = f_n - f_{n-1}$, $\nabla^2 f_n = \nabla f_n - \nabla f_{n-1} = f_n - f_{n-1} - (f_{n-1} - f_{n-2}) = f_n - 2f_{n-1} + f_{n-2}]$

$$f \simeq f_n + s\nabla f_n + \frac{s(s+1)}{2}\nabla^2 f_n + \dots + \frac{s(s+1)\cdots(s+k-2)}{(k-1)!}\nabla^{k-1} f_n$$

where $s = \frac{x - x_n}{h}$ ⇒ $dx = h\,ds$

$$f \simeq \sum_{m=0}^{k-1} (-1)^m \binom{-s}{m} \nabla^m f_n$$

$$\binom{-s}{2} = \frac{(-s)(-s-1)}{2} = \frac{s(s+1)}{2}$$

When $x = x_{n-l}$, $s = \frac{x_{n-l} - x_n}{h} = -\frac{(x_n - x_{n-l})}{h} = -\frac{lh}{h} = -l$

When $x = x_{n+1}$, $s = \frac{x_{n+1} - x_n}{h} = 1$

$$y(x_{n+1}) - y(x_{n-l}) \simeq \int_{-l}^{1}\left[\sum_{m=0}^{k-1} (-1)^m \binom{-s}{m} \nabla^m f_n\right] h\,ds$$

$$y(x_{n+1}) = y(x_{n-l}) + h\sum_{m=0}^{k-1} \int_{-l}^{1}\Big[(-1)^m \binom{-s}{m} \underbrace{\nabla^m f_n}_{\text{constant}}\Big] ds$$

Denote $\int_{-l}^{1} (-1)^m \binom{-s}{m} ds = \beta_m^{(l)}$

$$y(x_{n+1}) = y(x_{n-l}) + h\sum_{m=0}^{k-1} \beta_m^{(l)} \nabla^m f_n$$

$$\boxed{y_{n+1} = y_{n-l} + h\sum_{m=0}^{k-1} \beta_m^{(l)} \nabla^m f_n}$$

This is called Adam's k-step explicit formula

Methods corresponding to $l = 0$ are called Adams-Bashforth formula.

Methods corresponding to $l = 1$ are called Nystrom's formula.

Adams-Bashforth formula

k-step Adams-Bashforth formula is given by

$$\boxed{y_{n+1} = y_n + h\sum_{m=0}^{k-1} \beta_m^{(0)} \nabla^m f_n}$$

where $\beta_m^{(0)} = \int_0^1 (-1)^m \binom{-s}{m} ds$

$$\beta_0^{(0)} = \int_0^1 (-1)^0 \cancel{\binom{-s}{0}}^{\,1} ds = 1$$

$$\beta_1^{(0)} = \int_0^1 (-1)\binom{-s}{1} ds = \int_0^1 s\,ds = \frac{1}{2}$$

$$\beta_2^{(0)} = \int_0^1 (-1)^2 \binom{-s}{2} ds = \int_0^1 \frac{s(s+1)}{2} ds = \frac{s^3}{6} + \frac{s^2}{4}\Big|_0^1 = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

$$\beta_3^{(0)} = \int_0^1 (-1)^3 \binom{-s}{3} ds = \int_0^1 \frac{s(s+1)(s+2)}{6} = \frac{3}{8}$$

$$\beta_4^{(0)} = \int_0^1 (-1)^4 \binom{-s}{4} ds = \frac{251}{720}$$

Table of values of $\beta_m^{(0)}$ →

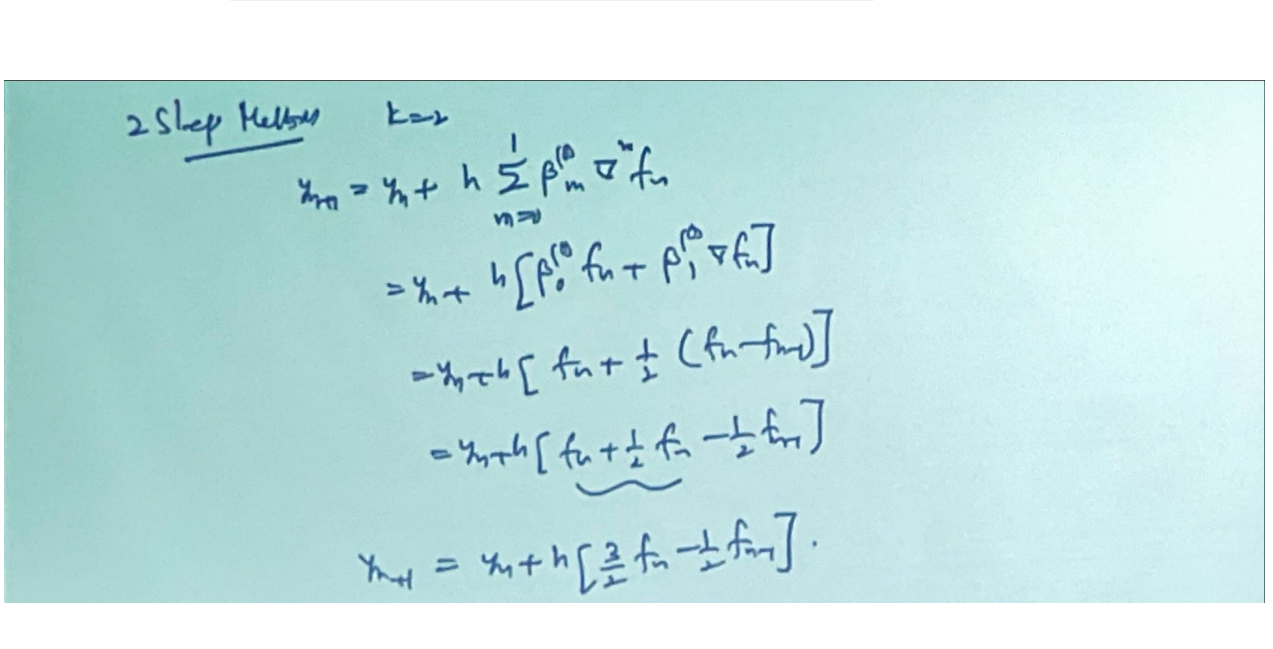
m	0	1	2	3	4	
$\beta_m^{(0)}$	1	1/2	5/12	3/8	251/720	....

Single (one) step method

$$y_{n+1} = y_n + h\sum_{m=0}^{0} \beta_m^{(0)} \nabla^m f_n$$

$$y_{n+1} = y_n + h\beta_0^{(0)} \nabla^0 f_n$$

$$\boxed{y_{n+1} = y_n + hf_n}$$ → Euler's method



2 Step Method k=2

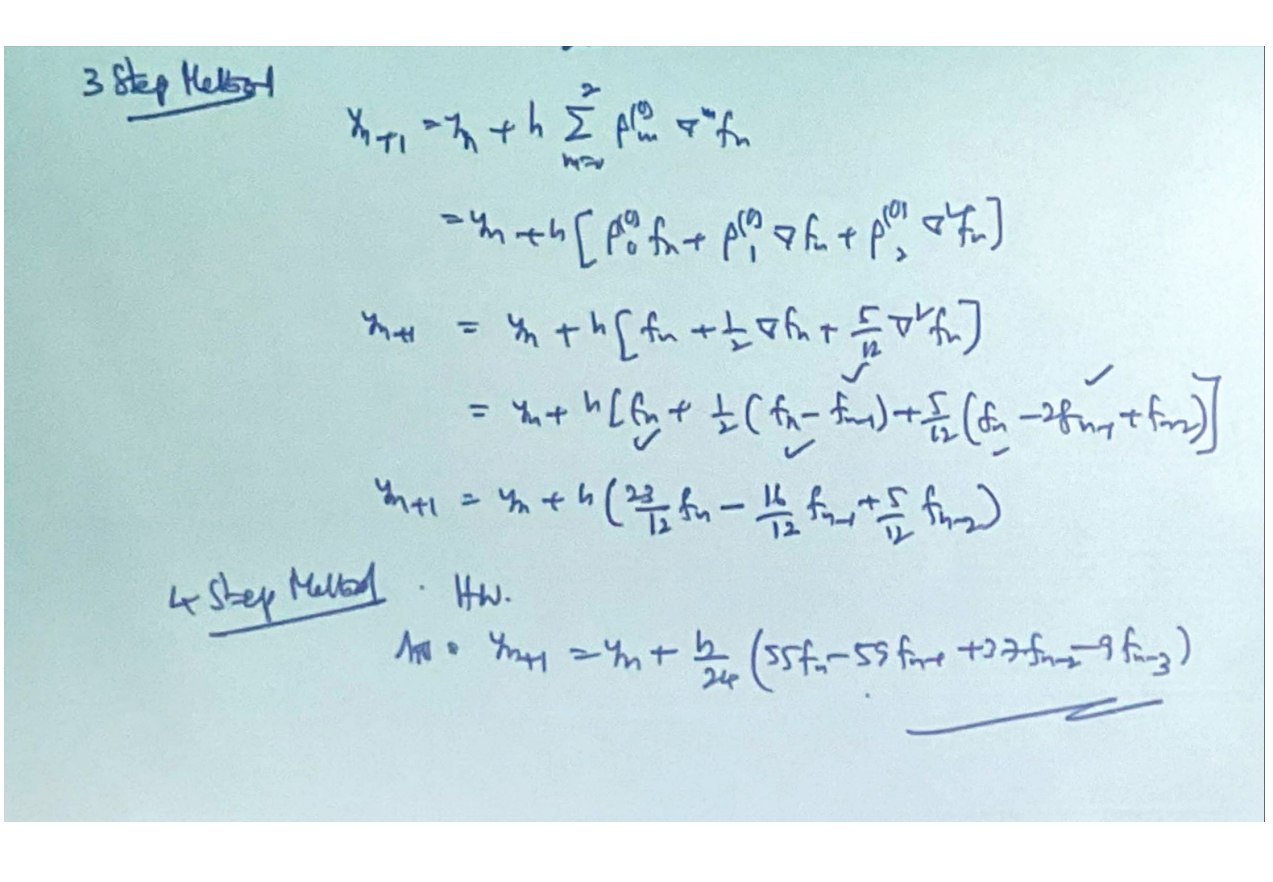
$y_{n+1} = y_n + h\sum_{m=0}^{1} \beta_m^{(0)} \nabla^m f_n$

$= y_n + h[\beta_0^{(0)} f_n + \beta_1^{(0)} \nabla f_n]$

$= y_n + h[f_n + \frac{1}{2}(f_n - f_{n-1})]$

$= y_n + h[f_n + \frac{1}{2} f_n - \frac{1}{2} f_{n-1}]$

$y_{n+1} = y_n + h[\frac{3}{2} f_n - \frac{1}{2} f_{n-1}]$.



3 Step Method

$y_{n+1} = y_n + h\sum_{m=0}^{2} \beta_m^{(0)} \nabla^m f_n$

$= y_n + h[\beta_0^{(0)} f_n + \beta_1^{(0)} \nabla f_n + \beta_2^{(0)} \nabla^2 f_n]$

$y_{n+1} = y_n + h[f_n + \frac{1}{2}\nabla f_n + \frac{5}{12}\nabla^2 f_n]$

$= y_n + h[f_n + \frac{1}{2}(f_n - f_{n-1}) + \frac{5}{12}(f_n - 2f_{n-1} + f_{n-2})]$

$y_{n+1} = y_n + h(\frac{23}{12} f_n - \frac{16}{12} f_{n-1} + \frac{5}{12} f_{n-2})$

4 Step Method · HW.

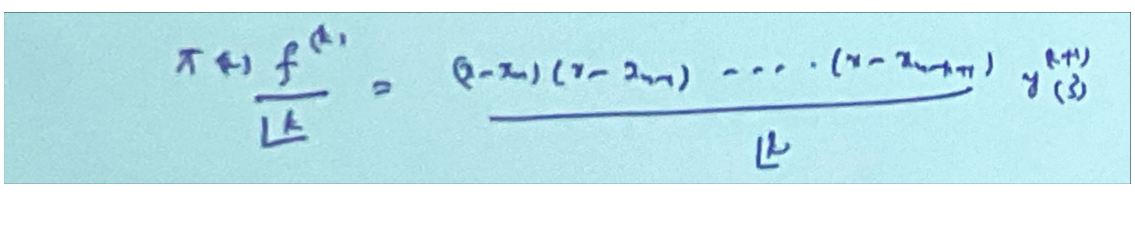
Ans: $y_{n+1} = y_n + \frac{h}{24}(55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3})$

What is the local truncation error for Adam's k-step method?

$$y_{n+1} = y_{n-l} + h\sum_{m=0}^{k-1} \beta_m^{(l)} \nabla^m f_n$$

where $\beta_m^{(l)} = \int_{-l}^{1} (-1)^m \binom{-s}{m} ds$

Error associated with the backward interpolating polynomial → (interpolating at $x_n, x_{n-1}, \dots, x_{n-k+1}$)



$$\frac{\pi(x) f^{(k)}}{k!} = \frac{(x-x_n)(x-x_{n-1})\cdots(x-x_{n-k+1})}{k!} y^{(k+1)}(\xi)$$

Local truncation error in Adam's k step method →

$$\int_{x_{n-l}}^{x_n} \frac{(x-x_n)\dots(x-x_{n-k+1})}{k!} y^{(k+1)}(\xi)\,dx$$

↳ function of x, ∴ depends on s as well.

Put $\frac{x - x_n}{h} = s$, $\frac{x - x_{n-i}}{h} = \frac{x - x_n + ih}{h} = s + i$

$$\therefore LTE = \int_{-l}^{1} \frac{sh(s+1)h\dots(s+k-1)h}{k!} y^{(k+1)}(\xi)\,h\,ds$$

$$= \int_{-l}^{1} \frac{s(s+1)\dots(s+k-1)}{k!} h^{k+1} y^{(k+1)}(\xi)\,ds$$

$$\boxed{LTE = \frac{h^{k+1}}{k!}\int_{-l}^{1} s(s+1)\dots(s+k-1)\, y^{(k+1)}(\xi)\,ds} \quad -①$$

$\to O(h^{k+1})$

∴ An explicit k-step method of Adam-type is of order k.

What is the error / LTE in the Adams-Bashforth type method?

LTE in k step Adams-Bashforth formula is obtained by putting $l = 0$ in ①.

$$LTE = h^{k+1}\int_0^1 \frac{s(s+1)\dots(s+k-1)}{k!} y^{(k+1)}(\xi)\,ds$$

$y^{(k+1)}(s)$ does not change sign in $(0,1)$. ∴ By 2nd mean value theorem, we get →

$$LTE = h^{k+1} y^{(k+1)}(\eta)\int_0^1 \frac{s(s+1)\dots(s+k-1)}{k!} ds \quad \text{for some } \eta \in (0,1)$$

* 2nd MVT ⇒ $\int_a^b f(x) g(x)\,dx = g(\delta)\int_a^b f(x)\,dx$ if g does not change sign in (a,b) for some $\delta \in (a,b)$

Nystrom formula

The k-step explicit methods corresponding to $l = 1$

A k step Nystrom's formula →

$$y_{n+1} = y_{n-1} + \sum_{m=0}^{k-1} \beta_m^{(1)} \nabla^m f_n$$

where $\beta_m^{(1)} = \int_{-1}^{1} (-1)^m \binom{-s}{m} ds$

$$= \int_{-1}^{1} \frac{s(s+1)\dots(s+m-1)}{m!} ds$$

$$\beta_0^{(1)} = \int_{-1}^{1} ds = 2$$

$$\beta_1^{(1)} = \int_{-1}^{1} s\,ds = 0$$

$$\beta_2^{(1)} = \int_{-1}^{1} \frac{s(s+1)}{2} ds = \int_{-1}^{1} \frac{s^2+s}{2} ds = \frac{1}{3}$$

$$\beta_3^{(1)} = \int_{-1}^{1} \frac{s(s+1)(s+2)}{6} ds = \int_{-1}^{1} \frac{s^3+3s^2+2s}{6} ds = \frac{1}{3}$$

$$\beta_4^{(1)} = \int_{-1}^{1} \frac{s(s+1)(s+2)(s+3)}{24} = \int_{-1}^{1} \frac{s^4+6s^3+11s^2+6s}{24} ds$$

$$= \frac{1}{24}\times\frac{2}{5} + \frac{11}{24}\times\frac{2}{3}$$

$$= \frac{1}{60} + \frac{11}{36} = \frac{29}{90}$$

Table of values of $\beta_m^{(1)}$ →

m	0	1	2	3	4	
$\beta_m^{(1)}$	2	0	1/3	1/3	29/90	.....

① $k = 1$ → $y_{n+1} = y_{n-1} + h\sum_{m=0}^{0} \beta_m^{(1)} \nabla^m f_n$

$$\boxed{y_{n+1} = y_{n-1} + 2hf_n}$$

Midpoint method obtained by central diff. approx.

This is a 2 step method (single-step Nystrom method tho)

② $k = 2$ →

$$\boxed{y_{n+1} = y_{n-1} + 2hf_n}$$

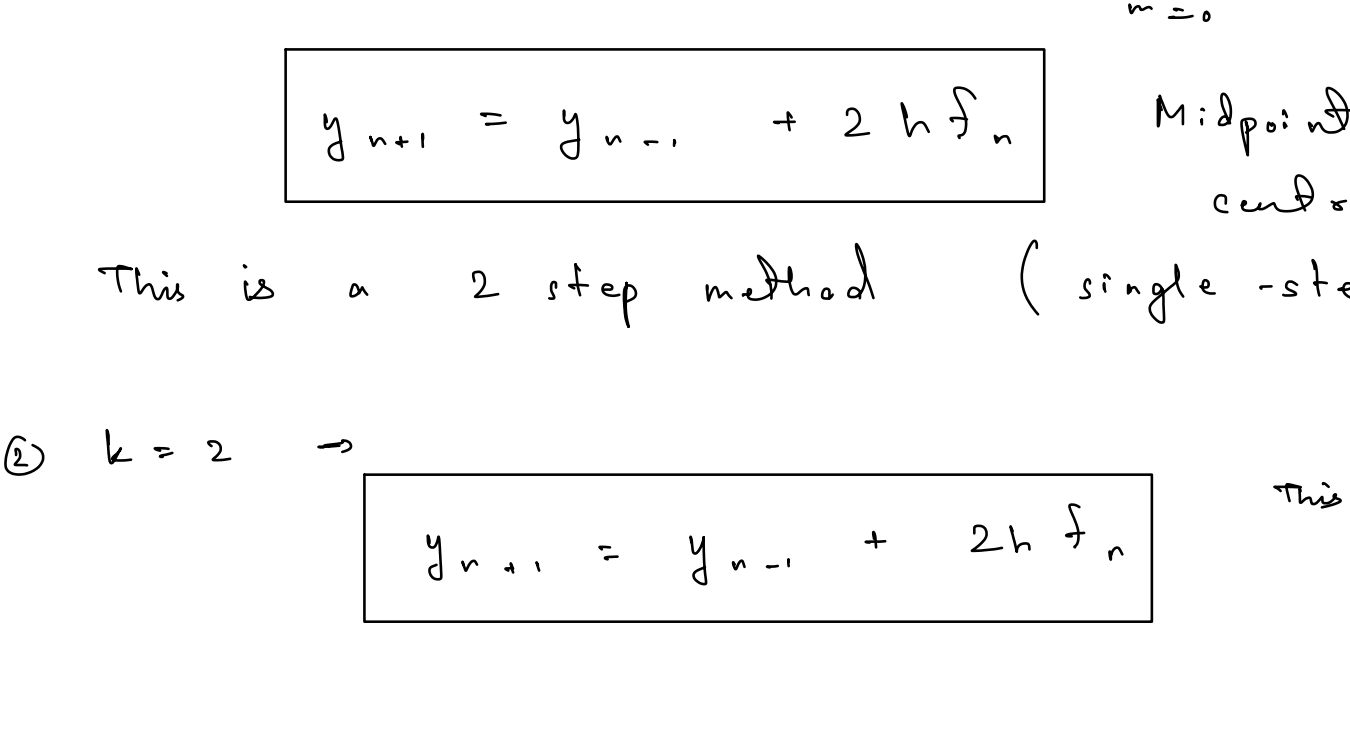
This is the same as $k=1$.

③ $k = 3$ →

$$y_{n+1} = y_{n-1} + 2hf_n + \frac{h}{3}\nabla^2 f_n$$

$$= y_{n-1} + 2hf_n + \frac{h}{3}(f_n - 2f_{n-1} + f_{n-2})$$

$$\boxed{y_{n+1} = y_{n-1} + h\left[\frac{7}{3} f_n - \frac{2}{3} f_{n-1} + \frac{1}{3} f_{n-2}\right]}$$



k=4 $y_{n+1} = y_{n-1} + h\sum_{m=0}^{3} \beta_m^{(1)} \nabla^m f_n$.

$= y_{n-1} + h[\beta_0^{(1)} f_n + \beta_1^{(1)} \nabla f_n + \beta_2^{(1)} \nabla^2 f_n + \beta_3^{(1)} \nabla^3 f_n]$

$= y_{n-1} + h[\frac{7}{3} f_n - \frac{2}{3} f_{n-1} + \frac{1}{3} f_{n-2} + \frac{1}{3}(f_n - 3f_{n-1} + 3f_{n-2} - f_{n-3})$

$= y_{n-1} + h(\frac{8}{3} f_n - \frac{5}{3} f_{n-1} + \frac{4}{3} f_{n-2} - \frac{1}{3} f_{n-3})$

LTE for k-step Nystrom method →

$$h^{k+1}\int_{-1}^{1} \frac{s(s+1)\dots(s+k-1)}{k!} y^{(k+1)}(\xi)\,ds$$

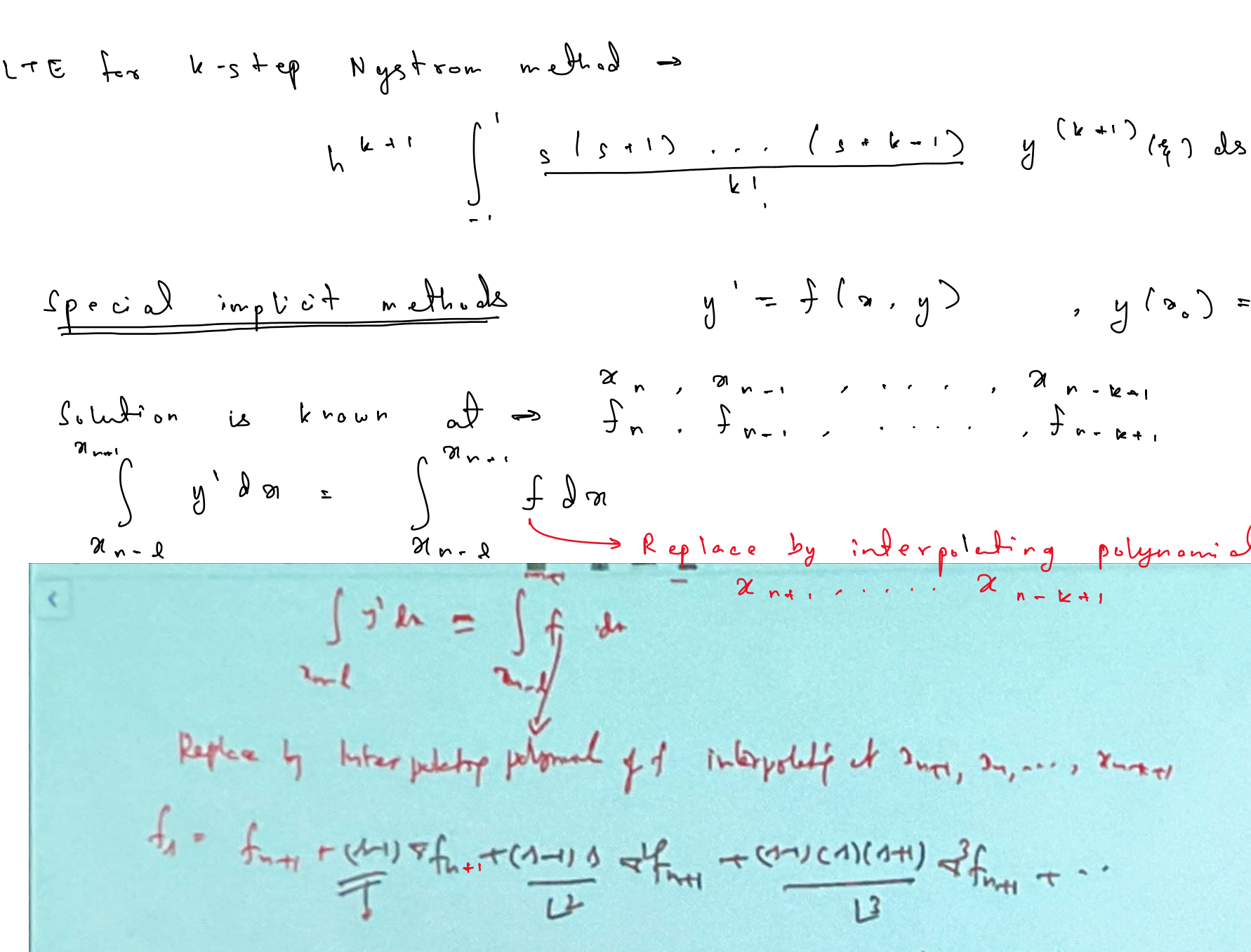
Special implicit methods

$y' = f(x,y)$, $y(x_0) = y_0$

Solution is known at → $x_n, x_{n-1}, \dots, x_{n-k+1}$; $f_n, f_{n-1}, \dots, f_{n-k+1}$

$$\int_{x_{n-l}}^{x_{n+1}} y'\,dx = \int_{x_{n-l}}^{x_{n+1}} f\,dx$$

→ Replace by interpolating polynomial at $x_{n+1}, \dots, x_{n-k+1}$



$\int_{x_{n-l}}^{x_{n+1}} y'\,dx = \int_{x_{n-l}}^{x_{n+1}} f\,dx$

Replace by interpolating polynomial of f interpolating at $x_{n+1}, x_n, \dots, x_{n-k+1}$

$f_s = f_{n+1} + \frac{(s-1)}{1!}\nabla f_{n+1} + \frac{(s-1)s}{2!}\nabla^2 f_{n+1} + \frac{(s-1)s(s+1)}{3!}\nabla^3 f_{n+1} + \cdots$

$\cdots + \frac{(s-1)(s)(s+1)\cdots(s+k-2)}{k!}\nabla^k f_{n+1}$

$s = \frac{x-x_n}{h}$.

$\frac{x-x_n}{h} = s$

$\frac{x - x_{n+1}}{h} = \frac{x - (x_n + h)}{h} = s - 1$

$$f_s = \sum_{m=0}^{k} (-1)^m \binom{1-s}{m} \nabla^m f_{n+1} \quad -①$$

Replacing on the RHS by its interpolating polynomial f_s given by ①,

$$y(x_{n+1}) - y(x_{n-l}) = \int_{-l}^{1} h\sum_{m=0}^{k} (-1)^m \binom{1-s}{m} \nabla^m f_{n+1}\,ds$$

$$= h\sum_{m=0}^{k}\int_{-l}^{1} (-1)^m \binom{1-s}{m} \nabla^m f_{n+1}\,ds$$

$$= h\sum_{m=0}^{k} \gamma_m^{(l)} \nabla^m f_{n+1}$$

$$y(x_{n+1}) \simeq y(x_{n-l}) + h\sum_{m=0}^{k} \gamma_m^{(l)} \nabla^m f_{n+1}$$

$$\boxed{y_{n+1} = y_{n-l} + h\sum_{m=0}^{k} \gamma_m^{(l)} \nabla^m f_{n+1}} \quad -(*)$$

(Denoting approx. for $y(x_{n+1})$ by y_{n+1})

where $\gamma_m^{(l)} = \int_{-l}^{1} (-1)^m \binom{1-s}{m} ds$

Adams special implicit method →

* Particular case → The formula (*) with $l = 0$ is called Adam-Moultons k step method and with $l = 1$ is called Milne's k-step method.

Adam-Moultons method

A k step method of this type is →

$$y_{n+1} = y_n + h\sum_{m=0}^{k} \gamma_m^{(0)} \nabla^m f_{n+1}$$

where $\gamma_m^{(0)} = \int_0^1 (-1)^m \binom{1-s}{m} ds$

$$\gamma_0^{(0)} = \int_0^1 ds = 1$$

$$\gamma_1^{(0)} = \int_0^1 (-1)(1-s)\,ds = \frac{1}{2} - 1 = -\frac{1}{2}$$

$$\gamma_2^{(0)} = \int_0^1 \frac{(1-s)(1-s-1)}{2} ds = \int_0^1 \frac{s^2-s}{2} = -\frac{1}{12}$$

$$\gamma_3^{(0)} = \int_0^1 (-1)\frac{(1-s)(1-s-1)(1-s-2)}{6} = \int_0^1 \frac{s^3-s}{6} = -\frac{1}{24}$$

$$\gamma_4^{(0)} = \frac{-19}{720} \qquad \gamma_5^{(0)} = \frac{-3}{180} \quad \text{and so on}$$

Table of coeff. → $(\gamma_m^{(0)})$

m	0	1	2	3	4	5	
$\gamma_m^{(0)}$	1	−1/2	−1/12	−1/24	−19/720	−3/180	...

k = 1

$$y_{n+1} = y_n + h(\gamma_0^{(0)} f_{n+1} + \gamma_1^{(0)} \nabla f_{n+1})$$

$$= y_n + h(f_{n+1} + (-\tfrac{1}{2})\nabla f_{n+1})$$

$$= y_n + h(f_{n+1} - \tfrac{1}{2}(f_{n+1} - f_n))$$

$$y_{n+1} = y_n + \frac{h}{2}(f_{n+1} + f_n)$$

This is trapezoid rule.

k = 2

$$y_{n+1} = y_n + h(\gamma_0^{(0)} f_{n+1} + \gamma_1^{(0)} \nabla f_{n+1} + \gamma_2^{(0)} \nabla^2 f_{n+1})$$

$$= y_n + h\left[f_{n+1} - \tfrac{1}{2}(f_{n+1} - f_n) - \tfrac{1}{12}(f_{n+1} - 2f_n + f_{n-1})\right]$$

$$= y_n + h\left[\frac{5}{12} f_{n+1} + \frac{8}{12} f_n - \frac{1}{12} f_{n-1}\right]$$

k = 3

$$y_{n+1} = y_n + h\left[\gamma_0^{(0)} f_{n+1} + \gamma_1^{(0)} \nabla f_{n+1} + \gamma_2^{(0)} \nabla^2 f_{n+1} + \gamma_3^{(0)} \nabla^3 f_{n+1}\right]$$

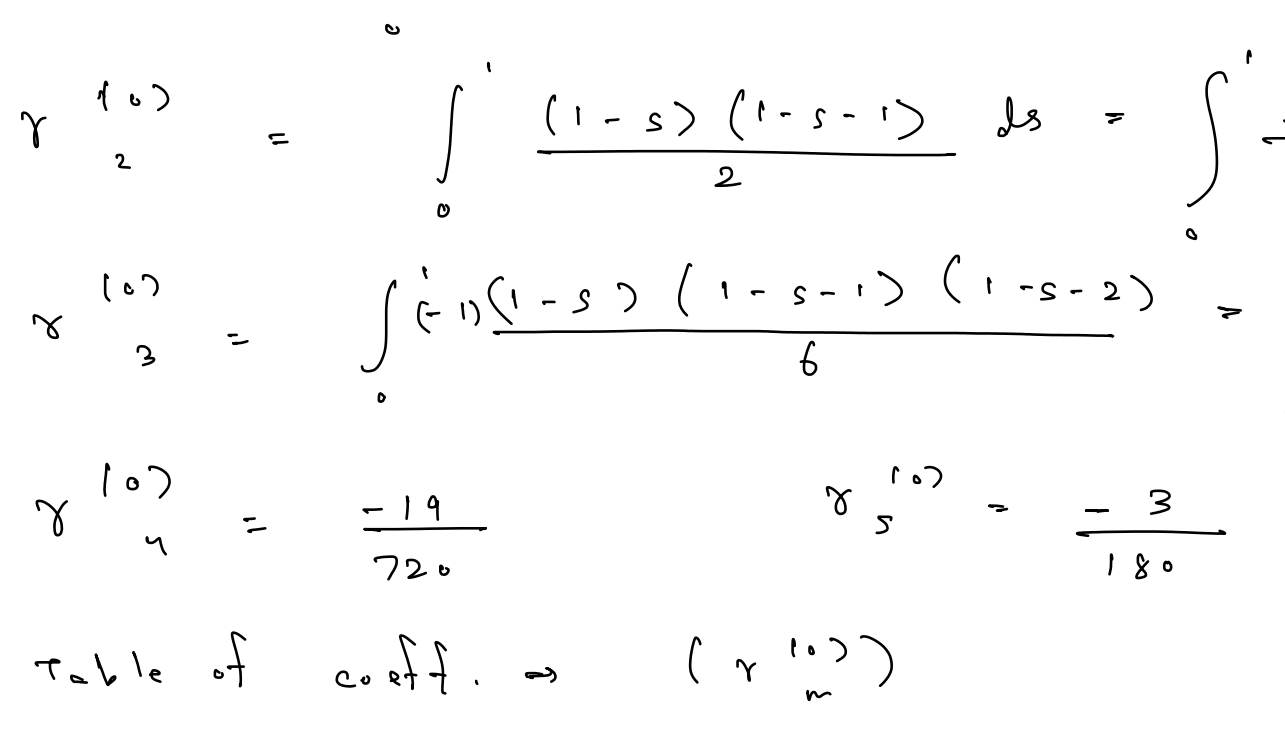
$$y_{n+1} = y_n + h\left[\frac{5}{12} f_{n+1} + \frac{8}{12} f_n - \frac{1}{12} f_{n-1} - \frac{1}{24}(f_{n+1} - 3f_n + 3f_{n-1} - f_{n-2})\right]$$

$$= y_n + h\left[\frac{9}{24} f_{n+1} + \frac{19}{24} f_n - \frac{5}{24} f_{n-1} + \frac{1}{24} f_{n-2}\right]$$

LTE for implicit multistep method ⇒

LTE formula ⇒

$$\int_{x_{n-l}}^{x_{n+1}} (x - x_{n+1})(x - x_n)\dots(x - x_{n-k+1}) \frac{f^{k+1}(\xi)}{(k+1)!}\,dx$$



LTE for Implicit Multistep Method

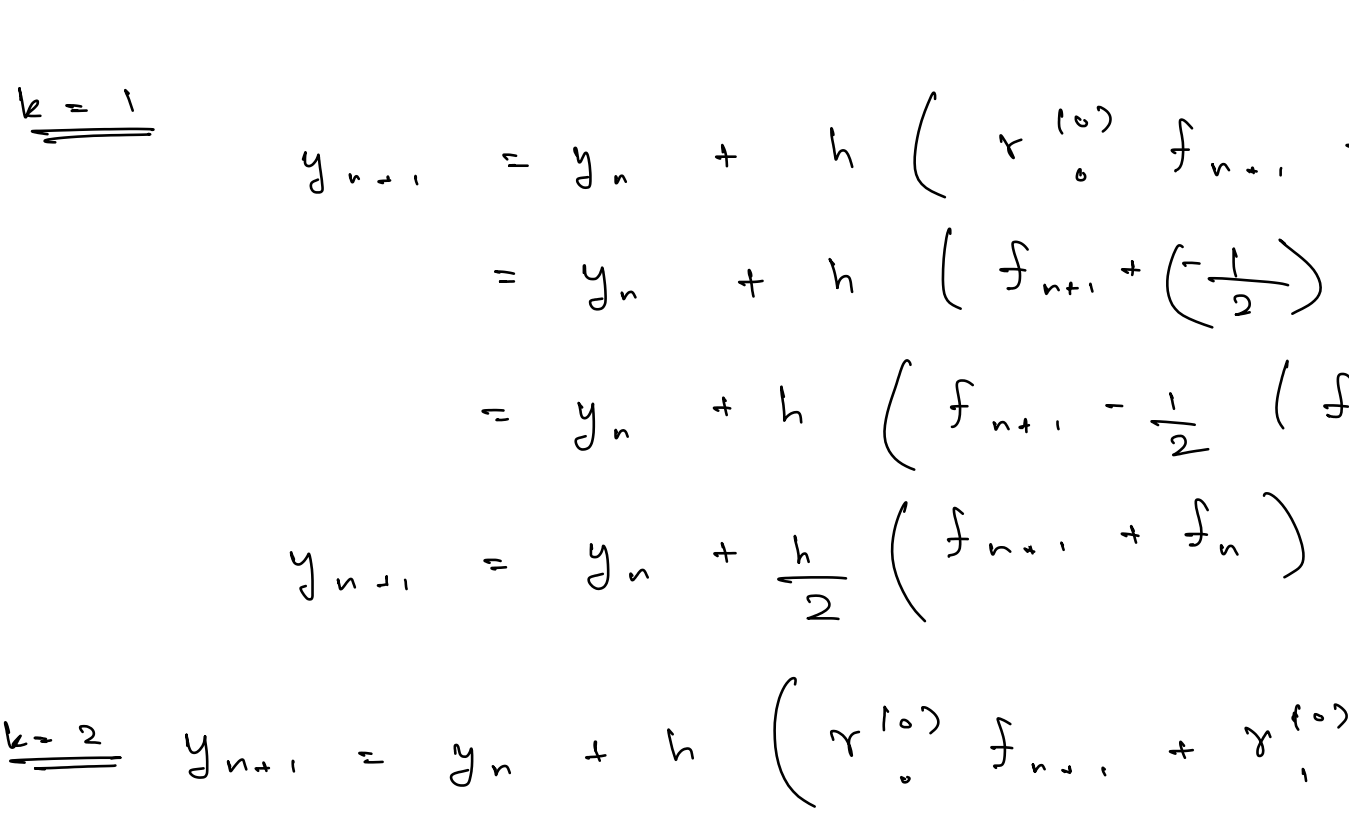
for the formula

$\int_{x_{n-l}}^{x_{n+1}} (x-x_{n+1})(x-x_n)\cdots(x-x_{n-k+1}) \frac{f^{(k+1)}(\xi)}{(k+1)!} dx$

Put $\frac{x-x_n}{h} = s$ and $dx = h\,ds$.

$LTE = \int_{-l}^{1} \frac{(s-1)h\,sh(s+1)h\cdots(s+k-1)h}{(k+1)!} f^{(k+1)}(\xi)\,h\,ds$

$= h^{k+2}\int_{-l}^{1} \frac{(s-1)s(s+1)\cdots(s+k-1)}{(k+1)!} y^{(k+2)}(\xi)\,ds$



LTE for Adams-Moultons k-step formula.

$l=0$

$LTE = h^{k+2}\int_0^1 \frac{(s-1)s(s+1)\cdots(s+k-1)}{(k+1)!} y^{(k+2)}(\xi)\,ds$

$= h^{k+2} y^{(k+2)}(\xi)\int_0^1 \frac{(s-1)s(s+1)\cdots(s+k-1)}{(k+1)!} ds$

(∵ $y^{(k+2)}(\xi)$ does NOT change sign in $(0,1)$)

$= h^{k+2}\,\gamma_{k+1}^{(0)}\,y^{(k+1)}(\eta)$

∴ k step Adams Moultons method is of order $k+1$.

Milne's formula

In the k step Adam implicit formula, put $l = 1$.

Milne's k step method is given by →

$$y_{n+1} = y_{n-1} + h\sum_{m=0}^{k} \gamma_m^{(1)} f_{n+1}$$

where $\gamma_m^{(1)} = \int_{-1}^{1} (-1)^m \binom{1-s}{m} ds$

$$\gamma_0^{(1)} = \int_{-1}^{1} ds = 2$$

$$\gamma_1^{(1)} = \int_{-1}^{1} (s-1)\,ds = -2$$

$$\gamma_2^{(1)} = \int_{-1}^{1} \frac{(s^2-s)}{2} ds = \frac{1}{3}$$

$$\gamma_3^{(1)} = \int_{-1}^{1} \frac{s^3-s}{6} ds = 0$$

$$\gamma_4^{(1)} = \int_{-1}^{1} \frac{(s^3-s)(s-2)}{24} ds = -\frac{1}{90}$$

Table of coeff. for $\gamma_m^{(1)}$ →

m	0	1	2	3	4	
$\gamma_m^{(1)}$	2	−2	1/3	0	−1/90	....

k = 1

$$y_{n+1} = y_n + h\left[\gamma_0^{(1)} f_{n+1} + \gamma_1^{(1)} \nabla f_{n+1}\right]$$

$$= y_n + h\left[2f_{n+1} - 2(f_{n+1} - f_n)\right]$$

$$= y_n + 2hf_n$$

k = 2

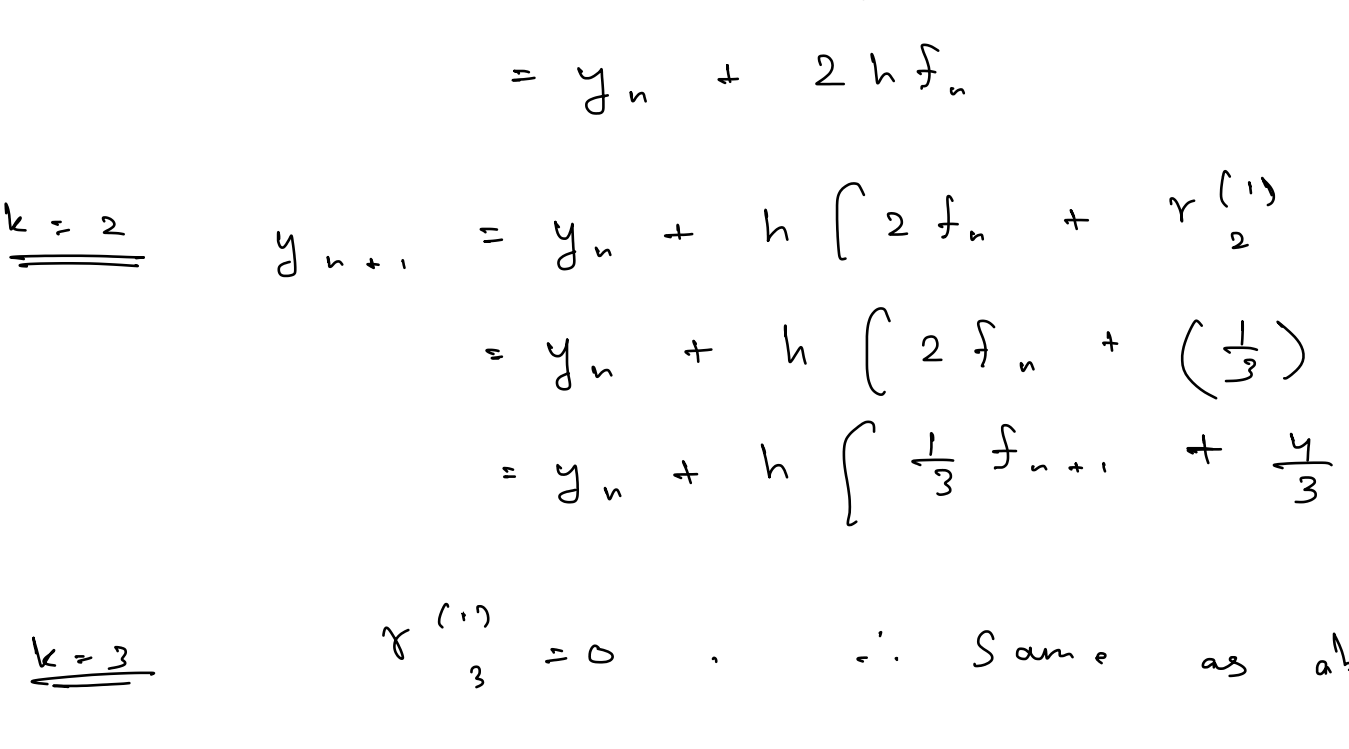
$$y_{n+1} = y_n + h\left[2f_n + \gamma_2^{(1)} \nabla^2 f_{n+1}\right]$$

$$= y_n + h\left[2f_n + (\tfrac{1}{3})(f_{n+1} - 2f_n + f_{n-1})\right]$$

$$= y_n + h\left[\frac{1}{3} f_{n+1} + \frac{4}{3} f_n + \frac{1}{3} f_{n-1}\right]$$

k = 3 $\gamma_3^{(1)} = 0$. ∴ Same as above

LTE of this method



LTE k-step Milne formula $\int_{-1}^{1} \frac{(s-1)s(s+1)\cdots(s+k-1)\,h^{k+2} f^{(k+1)}(\xi)}{(k+1)!} ds$

$= h^{k+2}\int_{-1}^{1} \frac{(s-1)s(s+1)\cdots(s+k-1)}{(k+1)!} y^{(k+2)}(\xi)\,ds$.

$O(h^{k+2})$.

∴ order of Milne's k step method is $k+1$.